



• car structure constraint: straight-line distance between sphere wheels stay constant $\Rightarrow r\dot{\varphi}_1 \sin(\alpha_1 + \beta_1) = r\dot{\varphi}_2 \sin(\alpha_2 + \beta_2)$
 $\Rightarrow \dot{\varphi}_1 \sin(\alpha_1 + \beta_1) + \dot{\varphi}_2 \sin(\alpha_2 + \beta_2) = 0 \quad \dots *$

• kinematic constraint:

passive castor wheel: constraints always satisfy.

powered spherical wheel:

$$[\sin(\alpha_1 + \beta_1) \quad -\cos(\alpha_1 + \beta_1) \quad -l \cos \beta_1] \dot{\mathbf{E}}_R = r \dot{\varphi}_1$$

$$[\cos(\alpha_1 + \beta_1) \quad \sin(\alpha_1 + \beta_1) \quad l \sin \beta_1] \dot{\mathbf{E}}_R = 0$$

$$[\sin(\alpha_2 + \beta_2) \quad -\cos(\alpha_2 + \beta_2) \quad -l \cos \beta_2] \dot{\mathbf{E}}_R = r \dot{\varphi}_2$$

$$[\cos(\alpha_2 + \beta_2) \quad \sin(\alpha_2 + \beta_2) \quad l \sin \beta_2] \dot{\mathbf{E}}_R = 0.$$

• controllable variable: $\beta_1, \beta_2, \dot{\varphi}_1, \dot{\varphi}_2$

if $\alpha_1 = 0, \alpha_2 = \lambda$, then:

$$\begin{bmatrix} \sin \beta_1 & -\cos \beta_1 & -l \cos \beta_1 \\ -\sin \beta_2 & \cos \beta_2 & -l \cos \beta_2 \\ \cos \beta_1 & \sin \beta_1 & l \sin \beta_1 \\ -\cos \beta_2 & -\sin \beta_2 & l \sin \beta_2 \end{bmatrix} R(\theta) \dot{\mathbf{E}}_2 = \begin{bmatrix} r \dot{\varphi}_1 \\ r \dot{\varphi}_2 \\ 0 \\ 0 \end{bmatrix} \quad \dots \begin{matrix} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{matrix}$$

$$\dot{\varphi}_1 \sin \beta_1 + \dot{\varphi}_2 \sin \beta_2 = 0 \quad \dots *$$

• Since car structure constrained the feasible domain of $[\beta_1, \beta_2, \dot{\varphi}_1, \dot{\varphi}_2]$.

so the kinematic constraints $\textcircled{1} \sim \textcircled{4}$ are not independent, any three of them can regulate the feedforward kinematics. Proof of dependence between $\textcircled{1} - \textcircled{4}$.

$\textcircled{1} + \textcircled{3} + * = \textcircled{4}$: assume $[-\cos \beta_2 \quad -\sin \beta_2 \quad l \sin \beta_2] R(\theta) \dot{\mathbf{E}}_2 = y \neq 0$

$\textcircled{1} \sin \beta_1 + \textcircled{3} \cos \beta_1$: $[1 \quad 0 \quad 0] R(\theta) \dot{\mathbf{E}}_2 = r \sin \beta_1 \dot{\varphi}_1$

$-\textcircled{2} \sin \beta_2 - \textcircled{4} \cos \beta_2$: $[1 \quad 0 \quad 0] R(\theta) \dot{\mathbf{E}}_2 = -r \sin \beta_2 \dot{\varphi}_2 - y \cos \beta_2$.

$\Rightarrow r \sin \beta_1 \dot{\varphi}_1 = -r \sin \beta_2 \dot{\varphi}_2 - y \cos \beta_2 \Rightarrow -y \cos \beta_2 = r \sin \beta_1 \dot{\varphi}_1 + r \sin \beta_2 \dot{\varphi}_2 = 0$

By randomness of β_2 , $y = 0$.

• $A_{3 \times 3} R(\theta) \dot{\mathbf{E}}_2 = \mathbf{b} \Rightarrow \dot{\mathbf{E}}_2 = R(\theta) A^{-1}$